HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Analyzed manufacturing operations and processes to identify opportunities for productivity gains, optimizing workflows across multiple cleanroom areas.
- Developed capacity and headcount models for operations, providing data-driven insights for resource allocation and future planning.
- Designed and implemented production standards across operations, contributing to enhanced process consistency and output quality.
- Mapped inventory flow and optimized material movement within and between manufacturing cells, reducing lead times by an average of **15%**.
- Applied visual factory concepts to production lines, improving real-time visibility into operational status and performance metrics.
- Collaborated with cross-functional engineering and operations teams to troubleshoot complex wafer-level processes and implement system enhancements.
- Containerized key Java Spring Boot services using **Docker** and deployed them to an **AWS EC2** environment, improving deployment consistency.

SKILLS

Manufacturing Operations – Process Analysis, Productivity Improvement, Capacity Modeling, Production Standards, Inventory Flow Optimization, Visual Factory Concepts, SEMI-E10, SEMI-E79

Software Development – Java, Spring Boot, REST APIs, SQL, PostgreSQL, Python, Docker, AWS

Data Analysis & BI – Power BI, Tableau, DAX, Power Query, Data Modeling, Data Warehousing

Automation & Scripting – Python Scripting, ETL, Shell Scripting

System Support & Operations – Production Support, Incident Response, SLA Management, Change Management, Linux

Project Management & Methodologies – Agile, Scrum, Jira, Confluence

PROJECTS

Manufacturing Process Optimization Initiative (HCL Technologies)

- Developed an Overall Manufacturing Efficiency model focused on SEMI-E10 and SEMI-E79 requirements, identifying key performance indicators for production.
- Analyzed complex multi-process cleanroom environments to pinpoint capacity limitations and propose targeted improvements, reducing bottlenecks by **20%**.
- Designed and documented new production standards based on efficiency analysis, facilitating a smoother rollout of process enhancements to the operations team.

Production Data Pipeline Automation (HCL Technologies)

- Engineered Python scripts to automate data extraction and transformation from operational databases, preparing datasets for analysis and reporting.
- Built scheduled ETL jobs using Python and SQL to consolidate data from various sources into a centralized repository, enabling faster ad-hoc querying by analysts.
- Implemented data validation checks within the pipeline, ensuring data integrity and reducing errors in downstream reporting by approximately **18%**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)